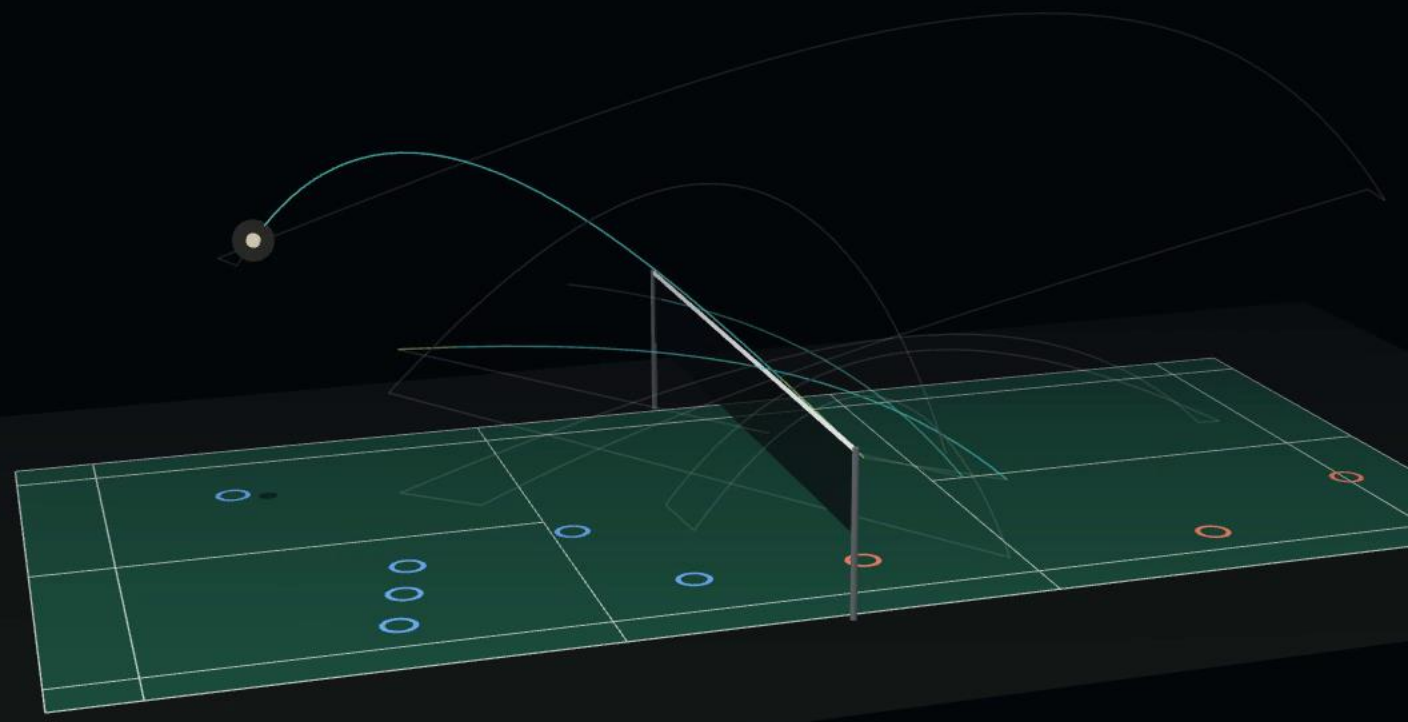
PRIVATE BADMINTON INTELLIGENCE

SHUTTLE COACH



A self-evolving team of AI coach agents, private and on-device

BATTLEFIELD THUNDERSTORM · INTEL ARC PRO B70 32GB · EVERYDAY AI

Team rm-rf-2 · github.com/gino6178/ASUS_hackathon

Pros get data-driven coaching. Everyone else guesses.

	Hire an analyst	Cloud analysis app	ShuttleCoach
Match video leaves the court	Shared with a person	Uploaded to a server	**Never: 0 bytes leave**
Time to a report	Days	Upload, then queue	**≤ match length**
Cost model	Salary or per-match fee	Monthly, per player	**One station + yearly update**
Works with no internet	Yes	No	**Yes**
3D flight and shot speed	Rarely	2D stats	**Every shot, metric 3D**
Gets better on its own	If the analyst studies	When the vendor updates	**Self-evolving: each coach agent evolves its own playbook**

Pro insight, priced for a club, and the footage never leaves the building.

Match video is personal data, so the analysis stays local



Biometric data

Faces, body shape and movement signatures, often of minors in school and junior programmes.



Tactical secrets

A team's footage is its game plan. Coaches will not upload it to a third-party cloud.



Courts are offline

Gyms and community halls have weak or no internet.

DESIGN RULE

Zero bytes of video or pose data leave the machine.

Every model, from shuttle tracking to the coaching agents, runs on one Intel Arc Pro B70.

Offline by default · no account · Telegram is opt-in and carries text only · one-step deletion · reports never share frames.

One camera in. A coaching staff out.

Record

Any fixed camera behind the court

Reconstruct

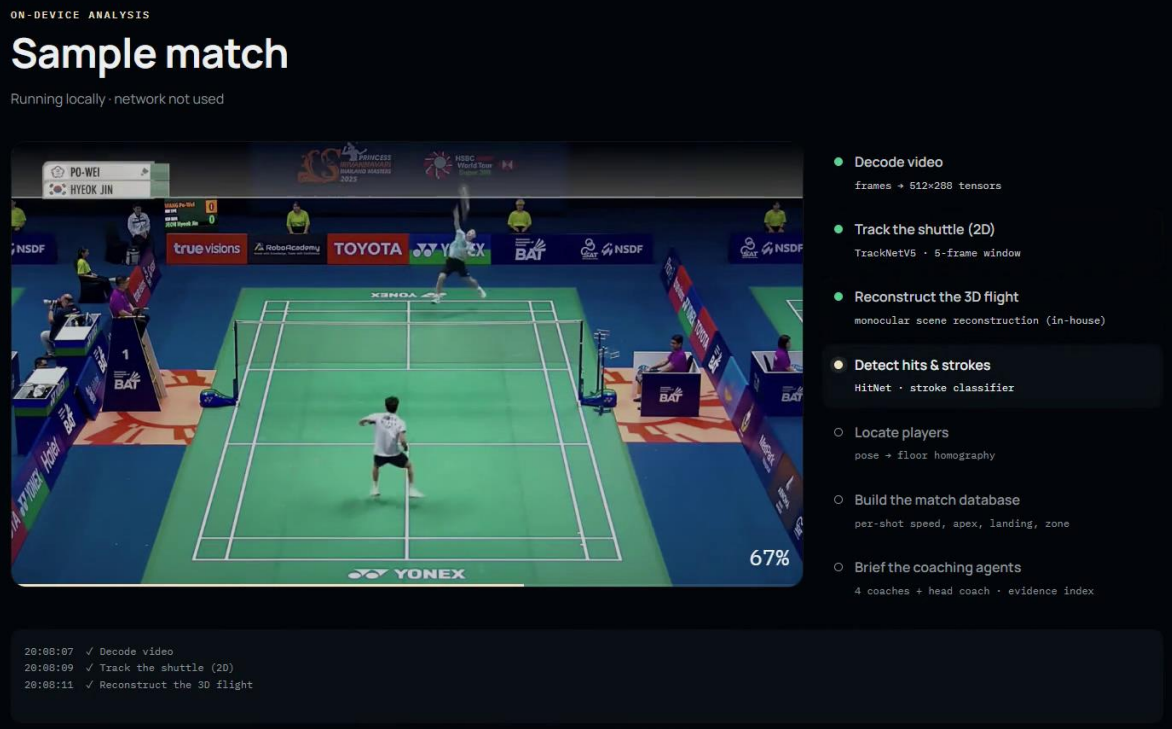
Shuttle flight in metric 3D, hits, player positions

Reason

Four self-evolving coach agents analyse, remember and simulate

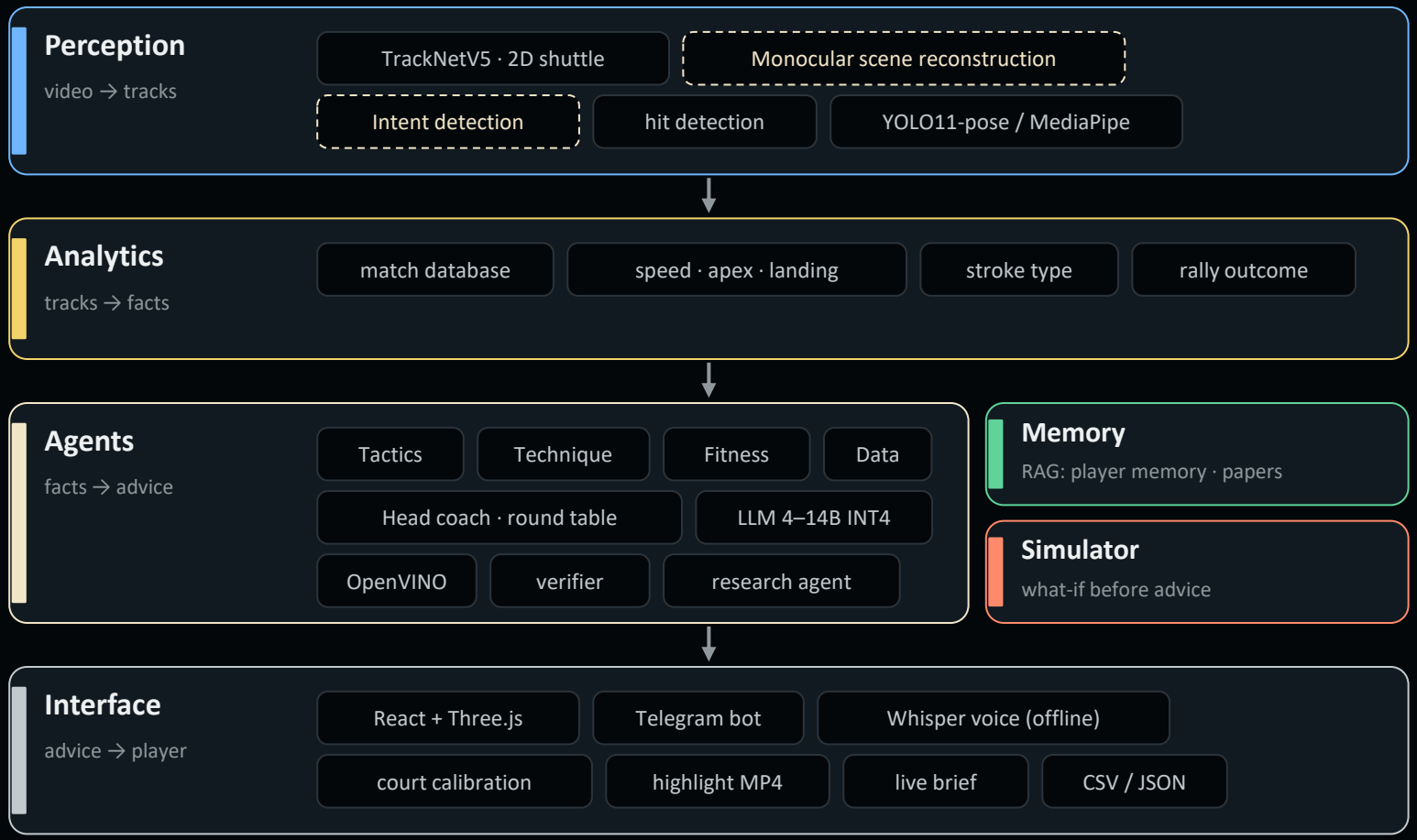
Act

Personified coaches flash zones, muscles and flights



Recorded from the prototype: the video is opened by the browser and never uploaded.

Solution: tools, technologies and methodology



Dashed chips: closed blocks. Everything runs on one local GPU.

METHODOLOGY

1 **Facts before language**

Deterministic code computes every number; the LLM only explains it.

2 **100% traceable**

A verifier recomputes every figure an agent says and links it to a rally, shot and video time.

3 **Simulate before advising**

Positioning changes must cut simulated losses first.

4 **Measure against ground truth**

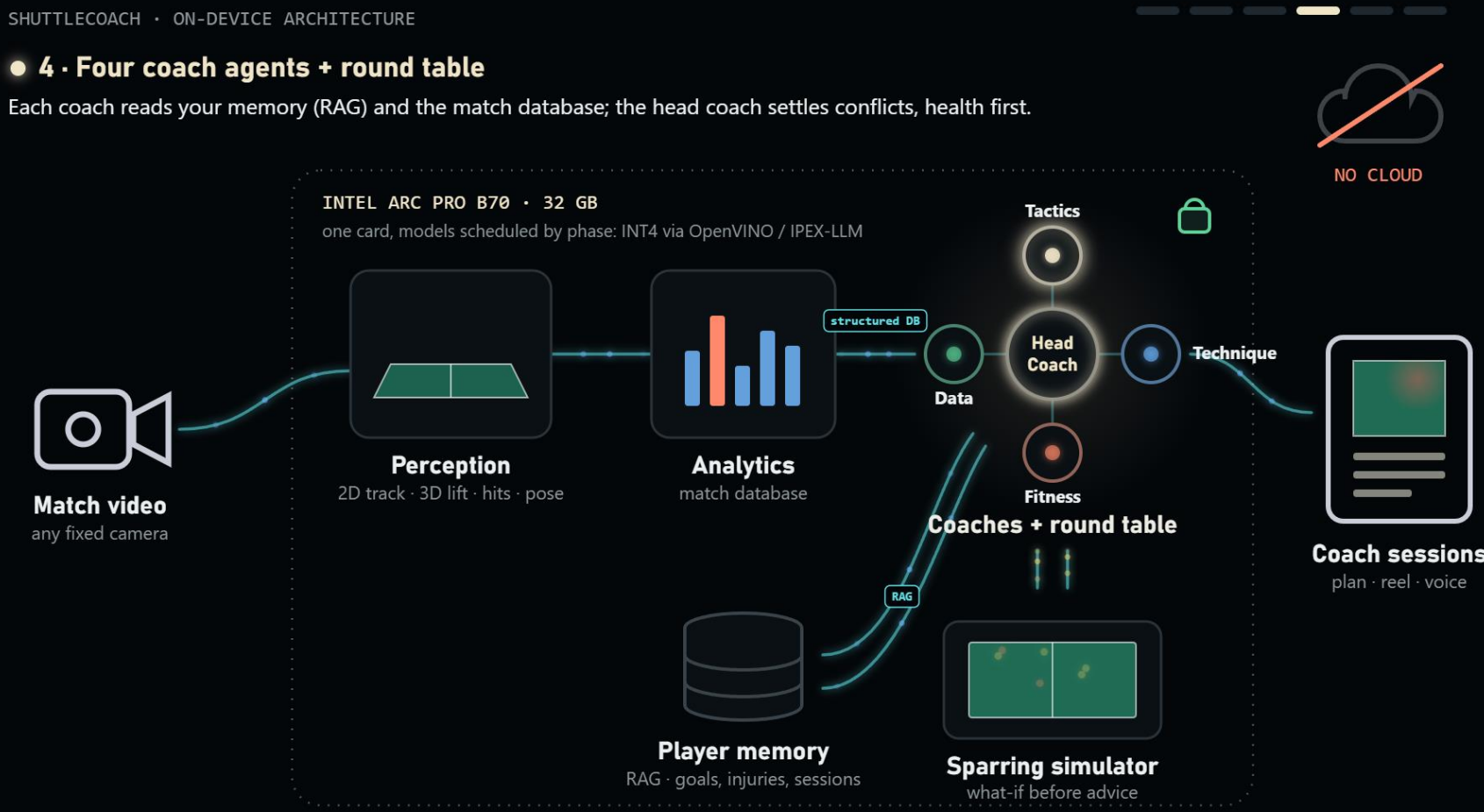
3D checked against a multi-camera capture; outcomes against labelled rallies.

How the system works, end to end

SHUTTLECOACH · ON-DEVICE ARCHITECTURE

4 · Four coach agents + round table

Each coach reads your memory (RAG) and the match database; the head coach settles conflicts, health first.



Every shot measured: 3D flight, speed, apex, contact height



2D TO 3D IN 3 STEPS

1 2D track. TrackNetV5 finds the shuttle in every broadcast frame.

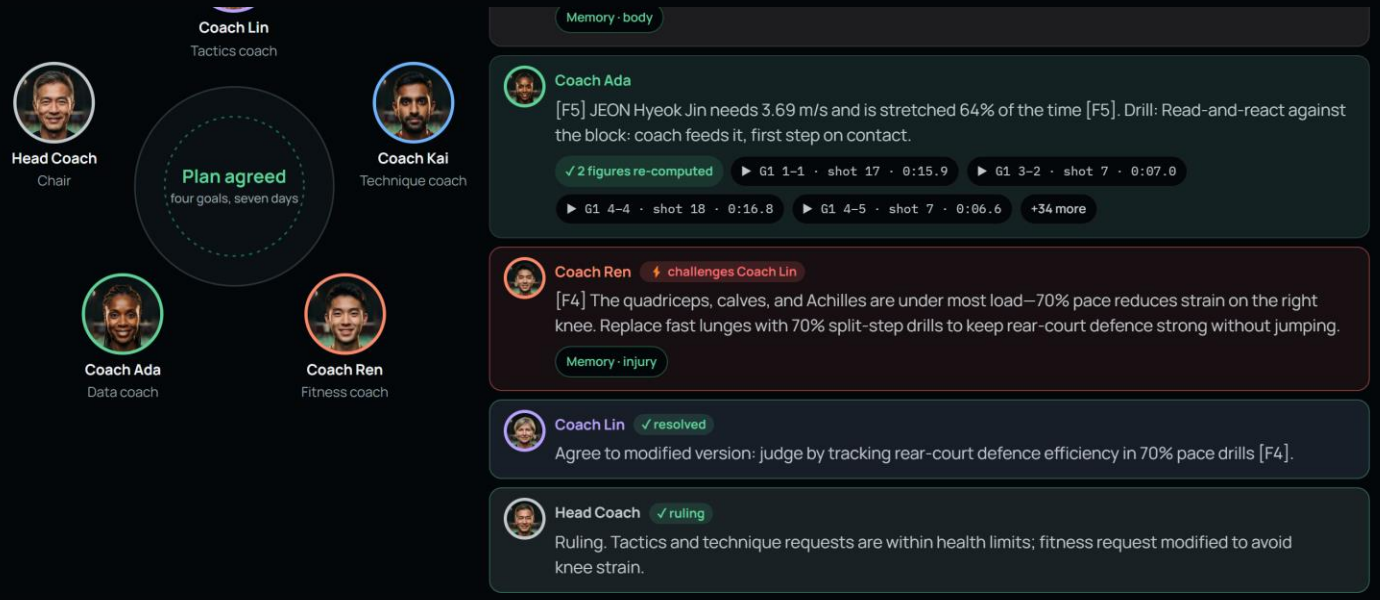
2 Camera, 3D and pose. Monocular scene reconstruction turns the 2D track into the flight in metres; pose gives both skeletons, speed and distance.

3 Speed curve. Each hit's launch speed, apex and contact height, clickable back to the frame.

200 km/h

far player's smash, measured from the 3D track

Round table: coaches who argue, then agree the week



HOW THE HEAD COACH RESOLVED THE CONFLICTS

1. **Health first.** Injuries and limits in the player's memory are hard constraints; no drill may break them.
2. **Then points at stake.** Requests are ranked by how many points that weakness cost in this match.
3. **One owner, one number.** Every kept request becomes a goal that the next recorded match re-measures.

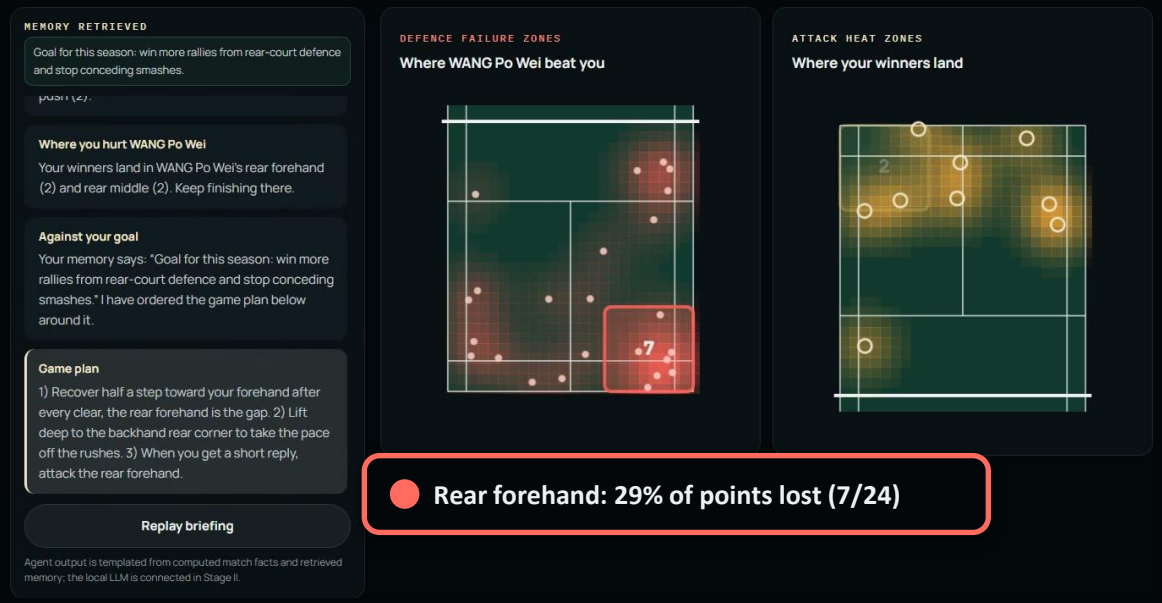
REQUEST	COACH	POINTS AT STAKE	MEMORY CHECK	RULING
Plyometric jump training	Coach Ren	**15** points lost in 15+ shot rallies	Blocked by memory: knee sore	Deferred: isometric strength instead
Net-margin work on net shots and lifts	Coach Kai	**14** own net errors	No conflict	Kept
Smash-defence blocks into the rear forehand	Coach Lin	**7** winners conceded there	Clashes with memory: knee sore, no split-step jumps	Kept, modified: 70% pace, low base
Read-and-react against the block	Coach Ada	**2** points ended by a block	No conflict	Kept

Written live by the agents
Each coach speaks on the local model from its own facts. Ren checks every request against memory (“right knee sore”) and challenges Lin's drill; Lin answers; the head coach rules: health first, then points at stake.

No invented numbers
Every turn passes the verifier (8 of 8 here), or falls back to the measured fact; evidence chips open the exact rally and shot.

Output
Four measurable goals and a 7-day schedule, written back to memory and sent to each player on Telegram.

Tactics and technique coaches: the zones and the flights





COACH LIN · TACTICS

Reads every point. 101 points: 9 winners hit vs 24 conceded; the match was won on 41 opponent errors.

Flashes the leak. Rear forehand: 7 of 24 points lost, blinking on the defence map.

Game plan. Recover toward the forehand after clears; lift deep to the backhand corner.

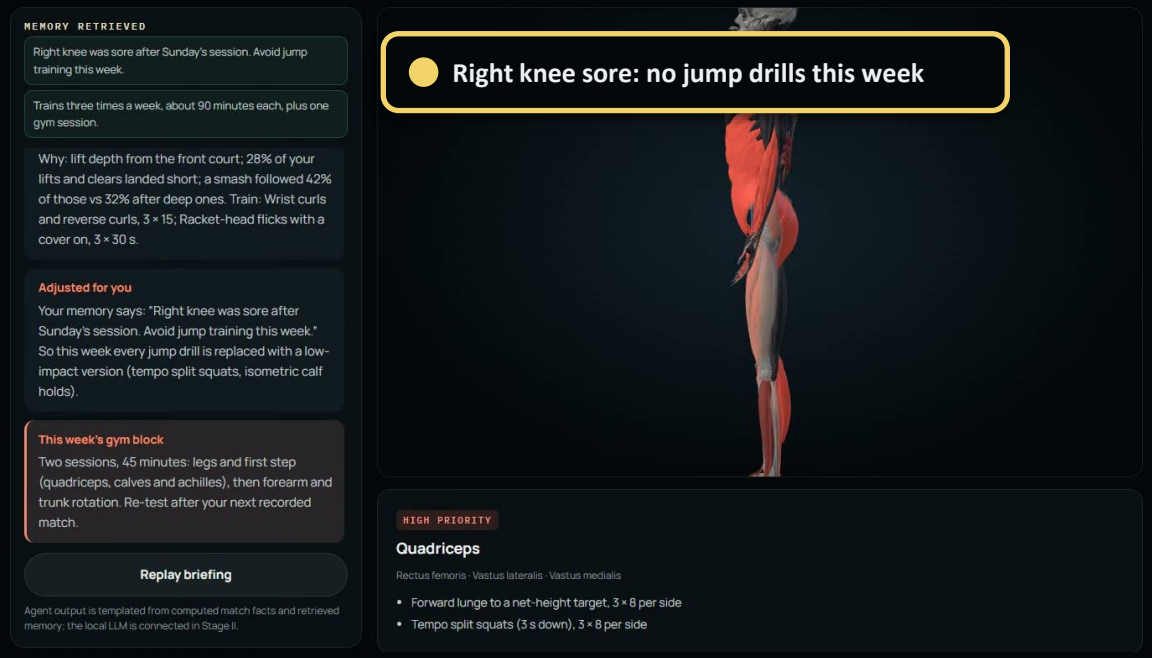
COACH KAI · TECHNIQUE

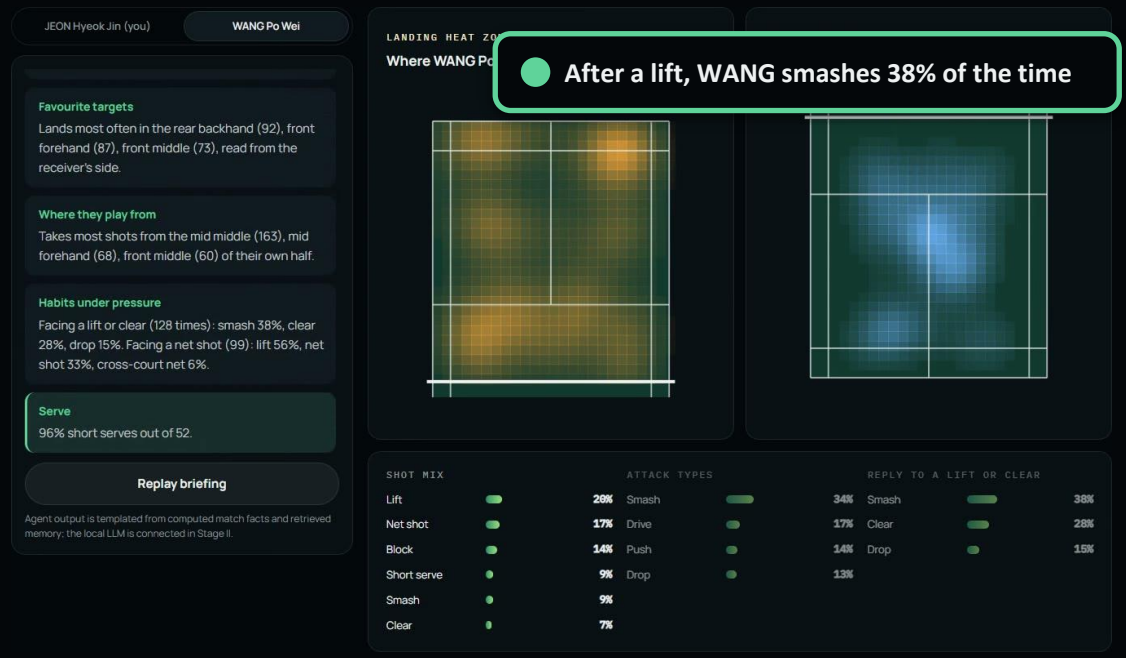
Every flight in 3D. Launch speed, apex, contact height, net clearance and depth per shot.

Shot quality grades. A deep 141 km/h clear earns an A; a net shot 40 cm over the tape a B.

What to fix. The weakest stroke goes first on next session's list.

Fitness and data coaches: the body and the habits





COACH REN · FITNESS

Muscles that flash. Quadriceps, calves, forearm and glutes pulse on a 402-muscle 3D model.

Why each one. Tied to findings: 60% of smashes into your half go to the mid forehand.

Adapts to memory. “Right knee sore”: jump drills become tempo split squats and isometric holds.

COACH ADA · DATA

Player profile. Shot mix, attack types and winners for either player, from every labelled shot.

Heat zones that blink. Favourite landing targets and hitting positions.

Habits under pressure. Facing a lift, the opponent smashes 38% of the time.

Self-evolving coach agents: each one evolves its own playbook

EVERY NIGHT, ON THE STATION

1 Research agent
Turns the player's weak points into paper searches; keeps a finding only if its numbers are in the abstract.

→

2 Knowledge base
Findings embedded on the station (RAG); each coach cites the closest studies as [E#].

→

3 Grounded reply
Match numbers only from facts [F#], study numbers only from [E#]; the verifier checks each one.

→

4 Feedback
Helpful / not helpful from players, plus every draft the verifier rejected.

→

5 Reflection
Each coach proposes at most two changes to its own playbook of rules.

↑ playbook v+1, used from the next reply

GATES, LIKE A HEAD COACH SIGNING OFF

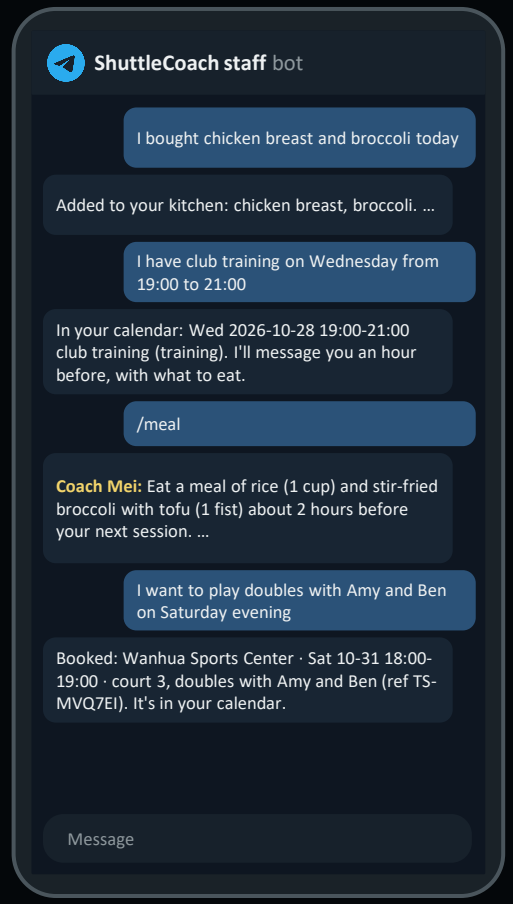
- Cites a study, in the coach's own area
- Passes the safety check; locked rules stay
- No regression on its test questions
- Health changes need a human coach; any version rolls back

First reflection run
A coach proposed lunges for knee pain, citing a real study. The gates held it for a human coach instead of adopting it.

67
findings from 24 papers, 3 nightly runs

0.8 s
median reply, local 4B

EXTENSION · 24-HOUR COACH



Bot output from its end-to-end test

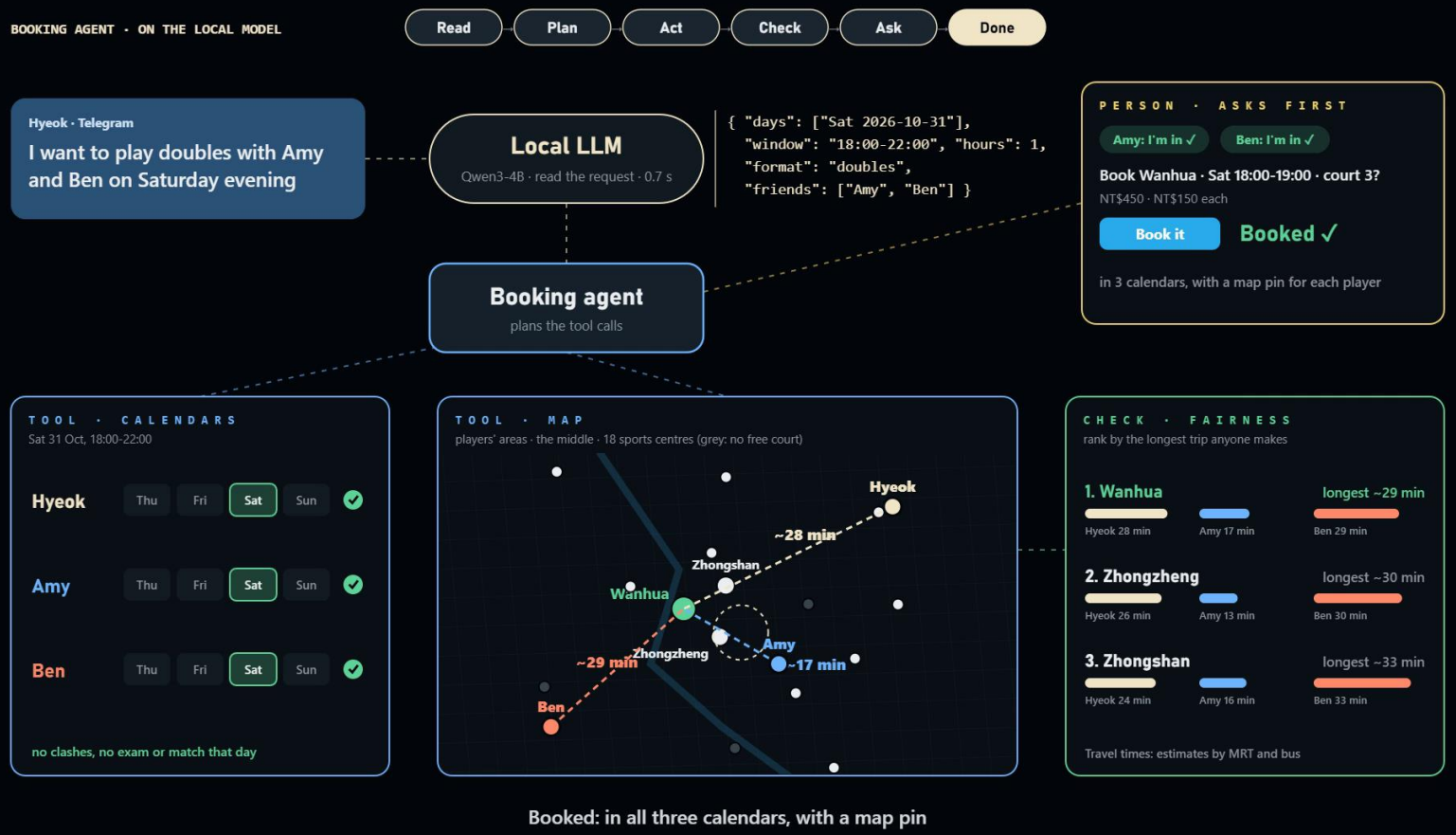
Food at home
meals timed to training

Schedule & log
reminder 1 h before; evening check-in

Court booking
a court in the middle; the agent books it on the venue's site after your OK

44 / 44
end-to-end checks passed

LLM agents that act: read, plan, use tools, check, ask



Booked: in all three calendars, with a map pin

EVERY AGENT RUNS THIS LOOP

1. **Read** — the local LLM turns a message into a structured request
2. **Plan** — picks the tools and fills their arguments
3. **Act** — map, calendars, memory, and its own browser on the venue's booking site
4. **Check** — verifier on every number; safety gates on advice
5. **Ask** — anything binding waits for a person's yes
6. **Learn** — feedback and nightly reflection

From a real run on the station. The agent then books on the venue's website in its own browser and stops before the final confirm.

Coach agents × 6
answer, advise, cite their evidence

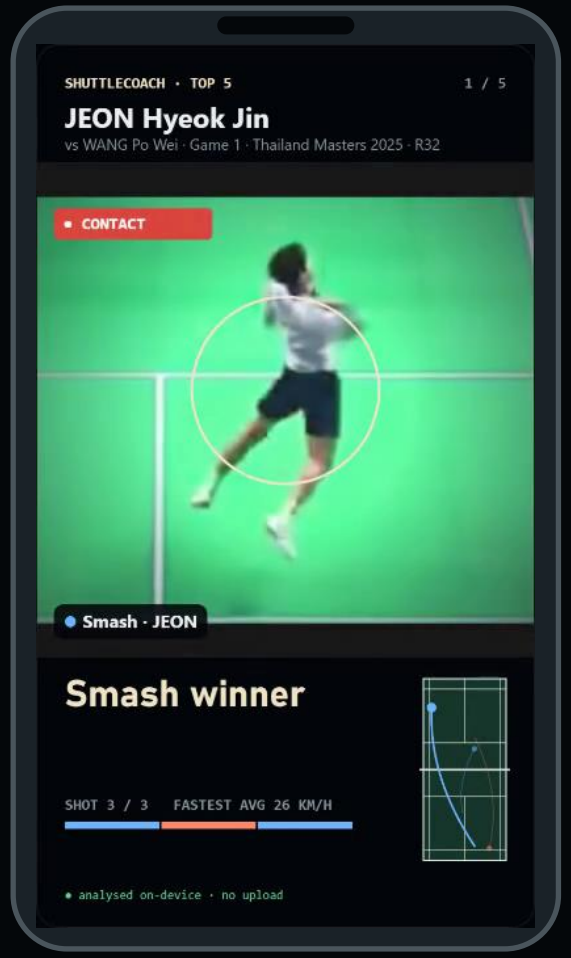
Booking agent
finds the fair court, books it on the venue's site

Research agent
searches and reads papers nightly

Reflection agent
proposes playbook changes behind gates

Reminder agent
before and after every session

Auto highlights: picked, filmed and shared on the device

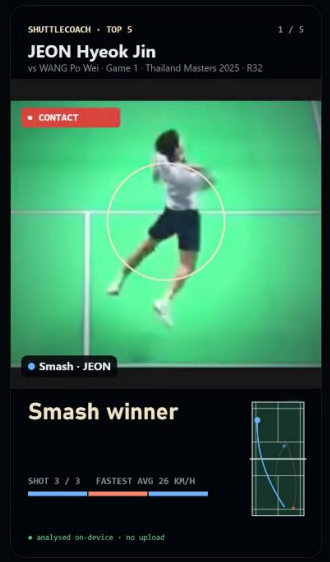


9:16, ready for Reels · TikTok · Shorts



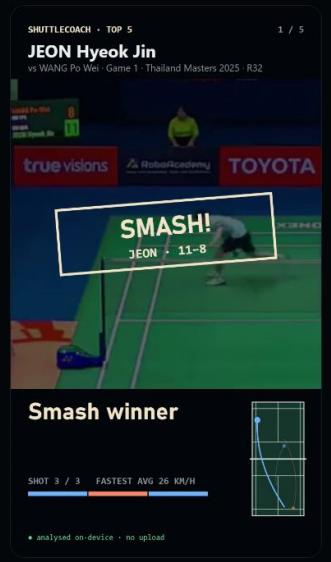
1 Follow-cam

tracks the shuttle between hitter and receiver



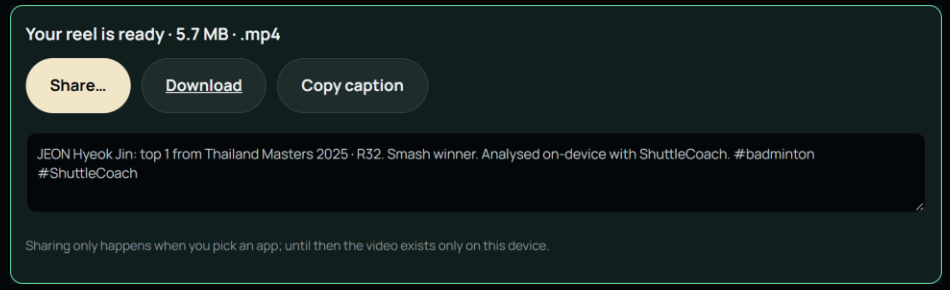
2 Slow-mo push-in

speed ramps down, camera closes on the hitter, holds on contact



3 Winner freeze

stamp, score and the winning shot

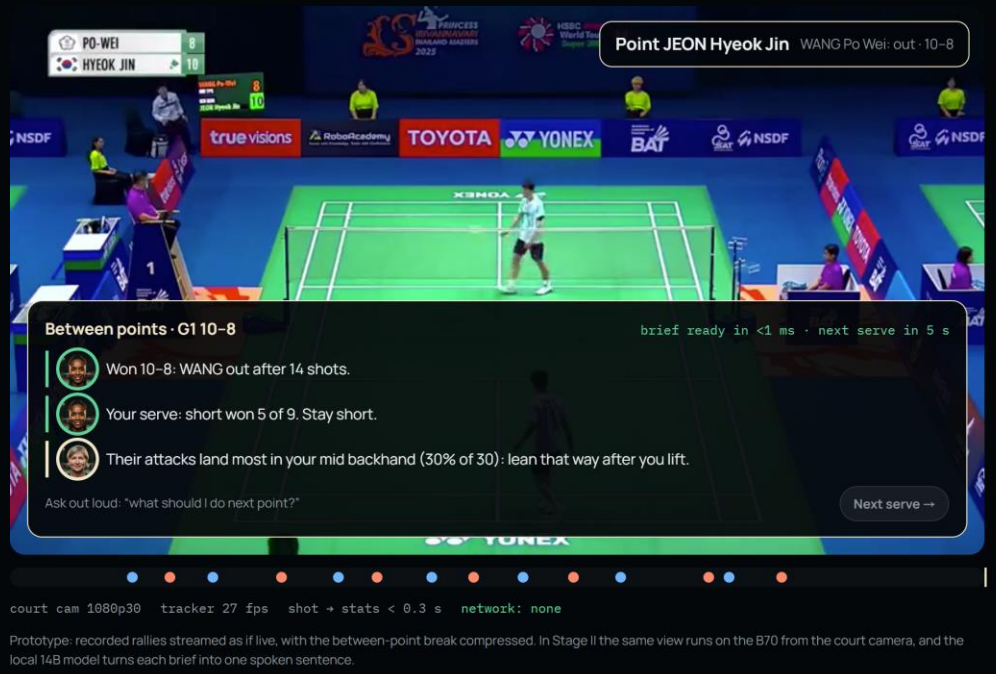


Picked automatically
Outright winners, rally length, smash finishes and pressure points (18+) rank every rally; untick one and the next best fills in.

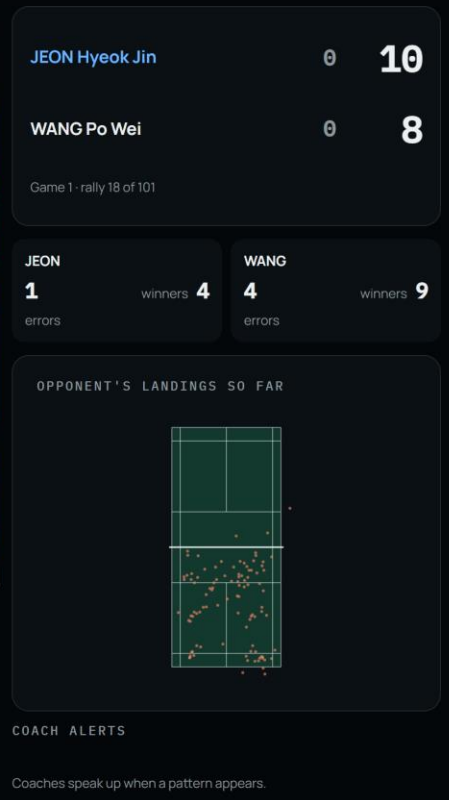
Ready to post
MP4 in 9:16 (Reels, TikTok, Shorts) or 16:9; one-tap Share, Download, and a written caption with hashtags.

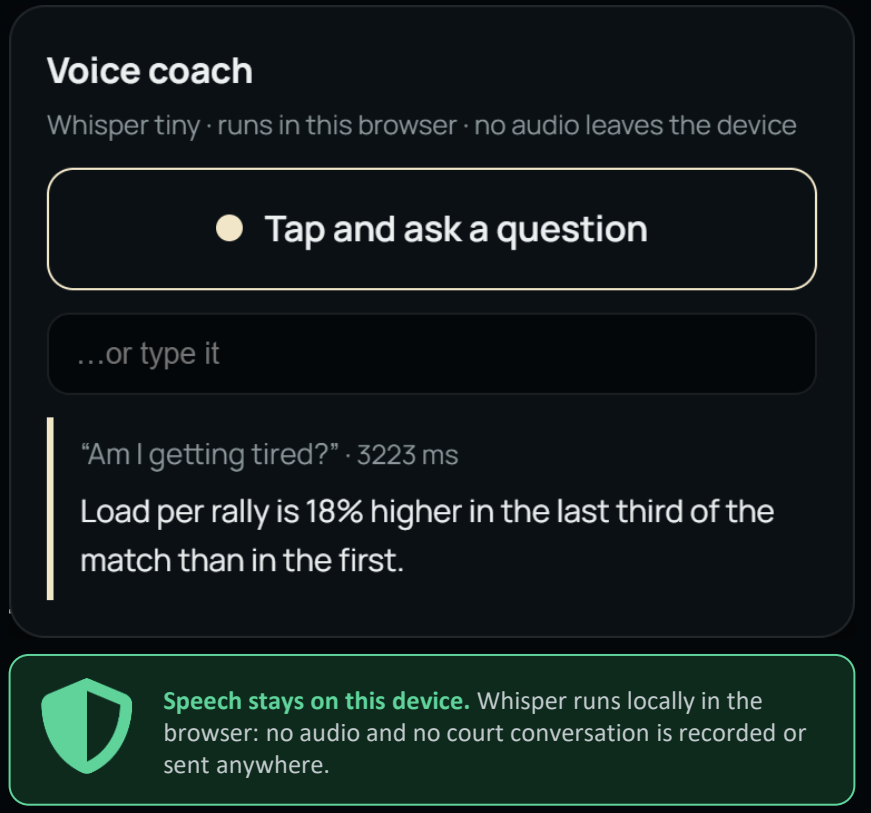
Why it sells
Players post their best rallies with the club's name on them: free marketing for the venue, and a reason to keep the subscription.

Live and hands-free: a brief in every break, answers by voice



Prototype: recorded rallies streamed as if live, with the between-point break compressed. In Stage II the same view runs on the B70 from the court camera, and the local 14B model turns each brief into one spoken sentence.





Speech stays on this device. Whisper runs locally in the browser: no audio and no court conversation is recorded or sent anywhere.

Between points (~20 s)

Last point, next serve, the opponent's hot zone and an error warning; longer briefs at 11 points and between games. Computed in under 1 ms.

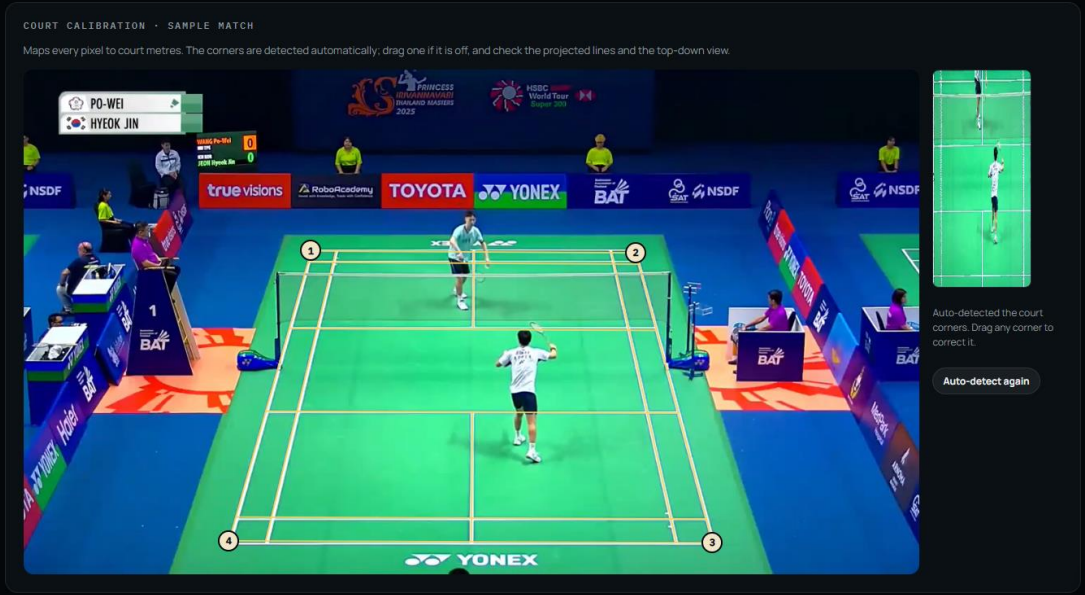
Hands-free voice

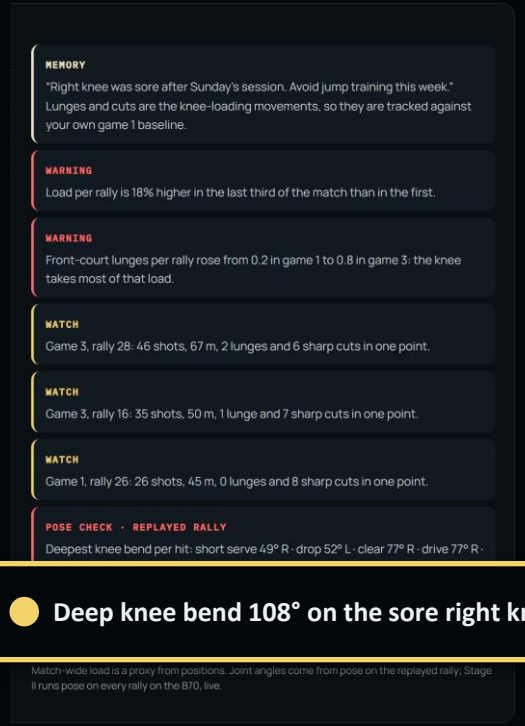
Tap Ask and speak: "What should I do next point?". Whisper tiny runs in the browser, so no audio leaves the court.

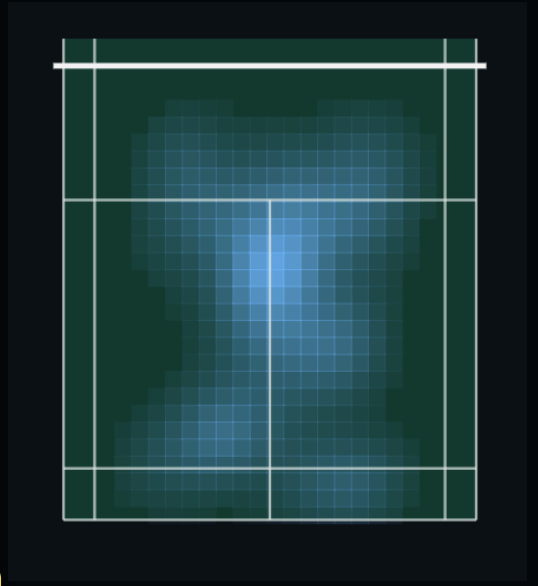
On the B70

Pose and tracking live from the court camera; a 4B model turns each brief into one spoken sentence in under a second; Whisper on OpenVINO.

Analyst toolkit: court calibration, injury watch, position maps







Court calibration

Corners found automatically from the court lines, draggable to correct; the court model is drawn back and a top-down view checks the fit.

Load and injury watch

Lunges and sharp cuts per rally from positions; pose skeletons give knee bend at every hit: 5 of 6 deep bends on the sore right knee.

Maps, scouting, export

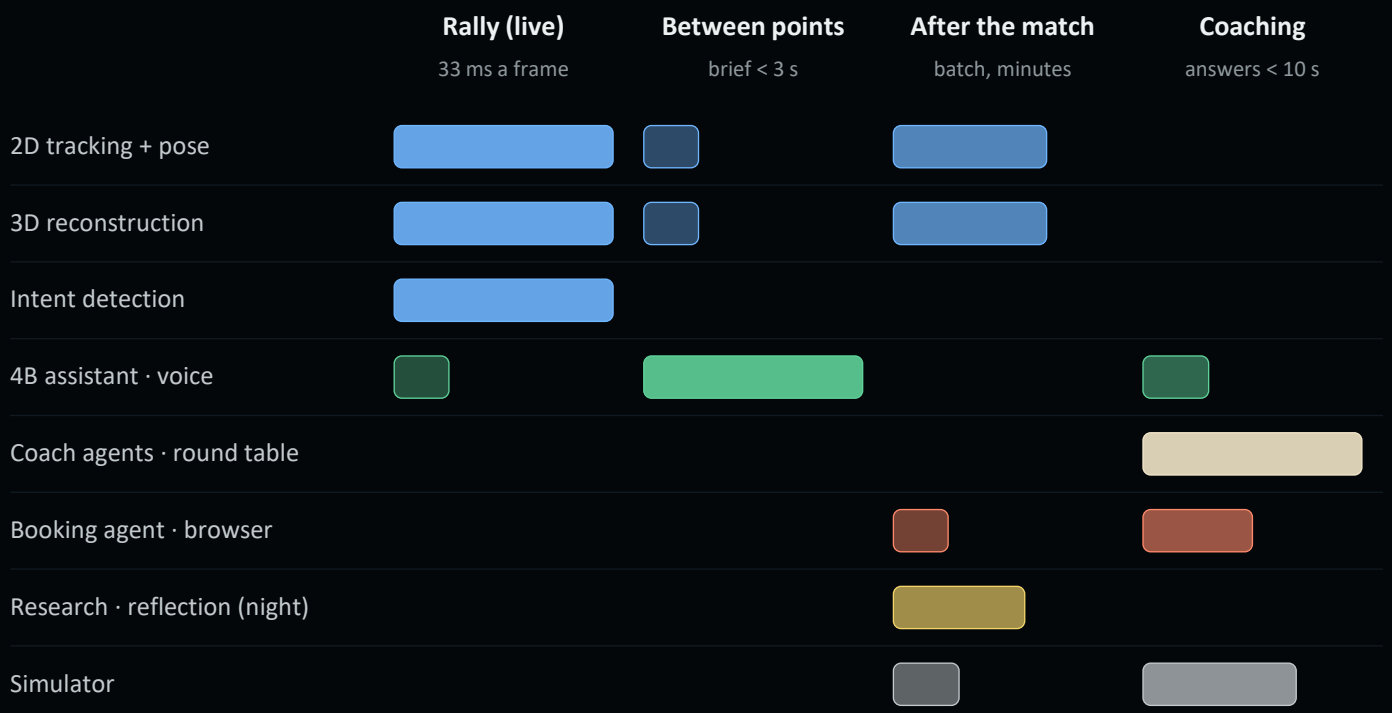
Where each player hits from (heat map or scatter); one-page scouting PDF; every shot as CSV or JSON.

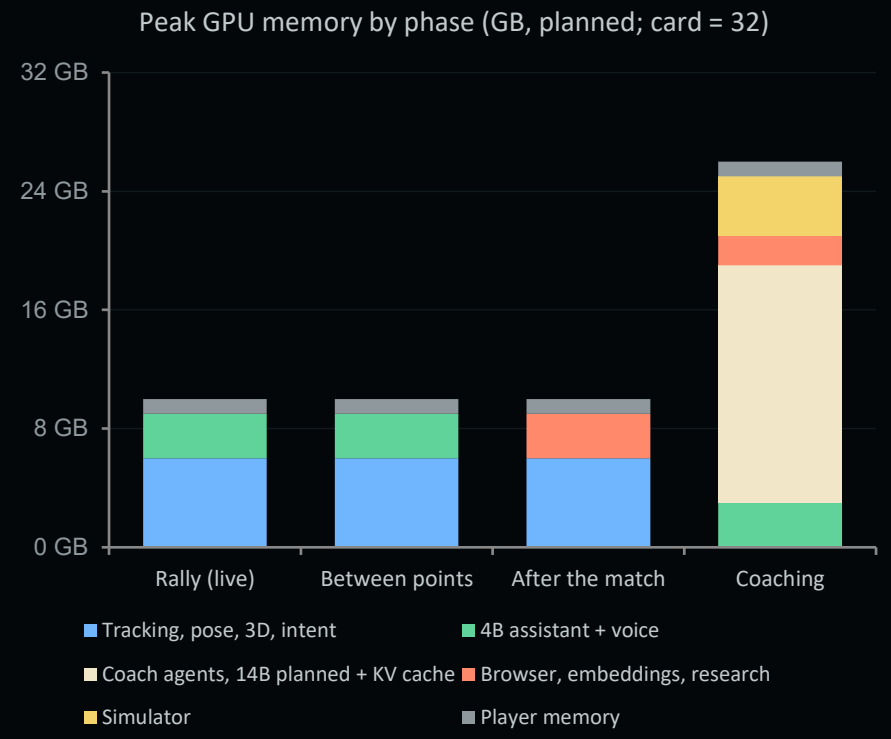
Deployment on ASUS UGen Intel Arc Pro B70



One GPU, scheduled by phase: what runs when

GPU TIME BY PHASE · BAR = SHARE OF THE CARD





Dynamic scheduler
A priority queue on one card: live perception keeps its 33 ms deadline, the between-point brief preempts batch work, post-match jobs and the nightly research and reflection fill idle time. **Today, on our test GPU, one 4B model serves every agent: coach chat, the round table, booking with its browser, research and reflection.**

Measured today on our test GPU (Qwen3-4B, 4-bit)
Coach reply 0.8 s median · round-table meeting 4.0 s for 9–10 verified turns · “I want to play” to a court offer 3.3 s, fitness coach included · booking agent in its browser: 28 steps, 9.4 s · tests 23/23 booking, 21/21 end to end. On the B70: INT4 ≈ 2.5 GB, decode ceiling about 240 tokens/s.

Expected benefits and outcomes

Players

- Where points are won and lost, by zone and stroke
- A weekly plan tied to their own footage
- Measurable progress per session

Coaches

- Hours of video review cut to minutes
- Advice backed by cited shots
- Test tactics in simulation first

Clubs and schools

- Analytics on every court, minors' video stays local
- One station, no per-upload fees
- A differentiated coaching service

Other sports

- Same pipeline: ball tracking, court calibration, 3D reconstruction, agents
- Tennis, table tennis, pickleball, volleyball
- New sport = new tracker + court model

NEXT DELIVERABLES

Working system

Upload a match video on the B70 and get the full report, end to end, offline

Open repository

Code, system architecture diagram, README, model information and performance data

Performance report

Speed, accuracy and latency measured on the B70; 3D error against reference data

Demo video

Real operation on the provided environment, 10 minutes, in English

Who pays, what it costs, and when it pays back

TAM
384 M people play badminton weekly worldwide (BWF, 2024); sports analytics US$2.3 B in 2025

SAM
≈ 2 M players in Taiwan: 9.4% of residents 13+ (Ministry of Sports, 2025); 113 commercial venues (MOF, May 2024), +16.5% a year

SOM
30 Stations year-3 target in Taiwan venues and school teams, about 1,800 subscribing players

B2B: club and venue

One Station shared by every member. 60 members on a NT$300/month “Pro analysis” add-on bring NT$18k a month against NT$7.5k a month of Station cost: hardware paid back in about 12 months, and a reason to join the club.

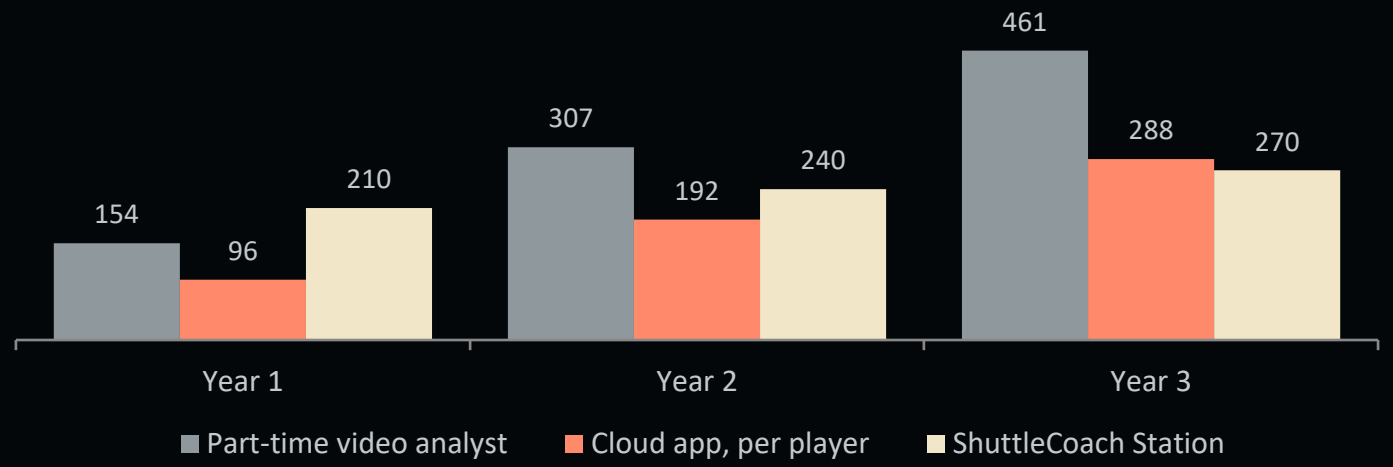
B2B2C: player subscription

NT$199/month through the club's Station: highlight reels, round-table plans, season tracking. Revenue split 70% club, 30% ShuttleCoach.

Schools and university teams

Same Station, no minors' video in the cloud; pilot with a university team and a local academy.

Cumulative cost for a 20-player club, NT$ thousand (illustrative)



1.5 years
payback against a part-time analyst

2.7 years
break-even against per-player cloud apps, with the video kept local

~12 months
club payback with 60 members on the NT$300 add-on

Illustrative assumptions: analyst 8 h/week at NT$400/h, 48 weeks; cloud NT$400/player/month; Station NT$180k + NT$30k/year (NT$7.5k/month over 3 years). Replace with quotes.

References and source code

SOURCE CODE https://github.com/gino6178/ASUS_hackathon

- TrackNetV3: shuttlecock tracking with augmentations and trajectory rectification. ACM MMAsia 2023. Code: MIT licence.
- TrackNetV5: robust shuttlecock tracking via motion prompts and spatiotemporal attentive fusion (backbone used frozen). Code: github.com/gino6178/TrackNetV5
- ShuttleSet: a human-annotated stroke-level singles dataset for badminton tactical analysis. KDD 2023.
- Monocular 3D reconstruction and intent detection: shown as closed blocks.
- 3D anatomy: BodyParts3D 4.0, © The Database Center for Life Science, CC BY 4.0; adapted via Human Atlas (MIT).
- Sample match: Thailand Masters 2025 R32 broadcast footage and hand annotations, used for a research prototype only.
- Libraries: Whisper tiny.en (MIT) via Transformers.js (Apache-2.0); MediaPipe Pose Landmarker (Apache-2.0). Feature ideas from Good-Badminton (yo-WASSUP): annotated video, court calibration, position maps.
- Agents: Qwen3-4B (Apache-2.0) served locally; OpenVINO GenAI path for the B70; Qwen3-Embedding for the knowledge base. Papers found through the OpenAlex, arXiv and Semantic Scholar APIs; messaging via the Telegram Bot API.
- Prototype status: coach agents answer on a local 4B model from computed match facts, every figure verified; tested on an RTX 3090 server, moving to the B70 next. Coach portraits are AI-generated, fictional people.

Team rm-rf-2